



**BAULKHAM HILLS
HIGH SCHOOL**

2020

**YEAR 11 YEARLY
EXAMINATION**

Mathematics Advanced

**General
Instructions**

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 11 – 31, show relevant mathematical reasoning and/or calculations

**Total marks:
73**

Section I – 10 marks (pages 2–5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 63 marks (Answer Booklet pages 4–16)

- Attempt Questions 11 – 31
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10

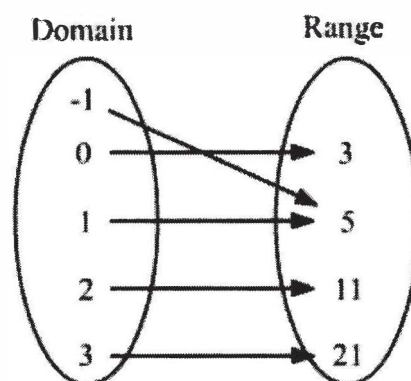
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

1 What is the value of $\sqrt{\frac{28-\pi^2}{5.3}}$ correct to 2 significant figures?

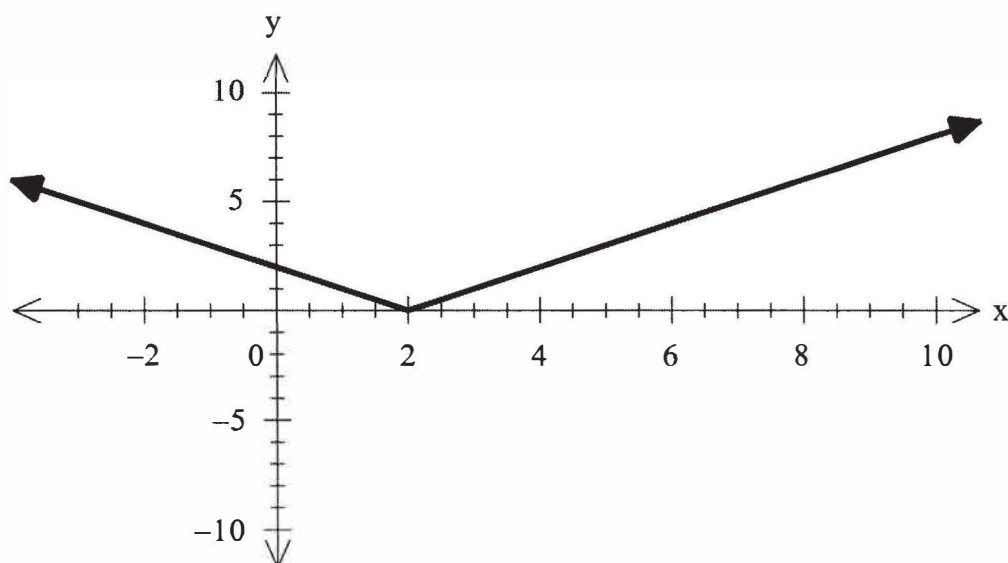
- A. 1.8
- B. 1.84
- C. 1.85
- D. 1.9

2 What type of relation is shown?



- A. One to one
- B. One to many
- C. Many to one
- D. Many to many

- 3 Which of the following statements is true about the following function?



- A. The function is neither continuous nor differentiable at $x = 2$.
- B. The function is continuous but not differentiable at $x = 2$.
- C. The function is differentiable but not continuous at $x = 2$.
- D. The function is both continuous and differentiable at $x = 2$.
- 4 Given that $\operatorname{cosec}\beta = -\frac{9}{5}$ and that $\tan\beta < 0$, what is the value of $\cos\beta$?

- A. $-\frac{9}{\sqrt{106}}$
- B. $-\frac{\sqrt{56}}{9}$
- C. $\frac{\sqrt{56}}{9}$
- D. $\frac{9}{\sqrt{106}}$

- 5 If $\log_b x = 3$ and $\log_b y = 5$, what is the value of $\log_b(xy^2)$?
- A. 8
- B. 13
- C. 28
- D. 75
- 6 What is the minimum value of the function $f(x) = 2x^2 - 4x + 1$?
- A. -2
- B. -1
- C. 1
- D. 2
- 7 What is the natural domain of $f(x) = \frac{1}{\sqrt{x^2 - 9}}$?
- A. $x < -3$ and $x > 3$
- B. $x \leq -3$ and $x \geq 3$
- C. $-3 < x < 3$
- D. $-3 \leq x \leq 3$

- 8 Which of the following is an equivalent expression for $\frac{(1+\cos x)(1-\cos x)}{\sin^2 x}$?
- A. $\tan^2 x$
 - B. $\cot^2 x$
 - C. $\operatorname{cosec}^2 x$
 - D. 1
- 9 If $f(x) = x^2 - 2x$ and $g(x) = \sqrt{x+1}$, what is $g(f(x))$?
- A. $(\sqrt{x+1})(x^2 - 2x)$
 - B. $x + 1 - 2\sqrt{x+1}$
 - C. $\sqrt{x^2 - x + 1}$
 - D. $|x - 1|$
- 10 Which of the following functions is **not** even?
- A. $f(x) = \sqrt{x^2 - 1}$
 - B. $f(x) = \frac{1}{1 + x^2}$
 - C. $f(x) = \operatorname{cosec} x$
 - D. $f(x) = \cos x$

End of Section I

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BAULKHAM HILLS HIGH SCHOOL
YEAR 11 Yearly Examination, 2020
MATHEMATICS ADVANCED

FINAL MARK

Name: _____

Teacher: _____

Question	F1	T1	S1	C1	E1	Total
MC	1,2,6,7,9,10 /6	4,8 /2		3 /1	5 /1	/10
Q.11-16 p 4-6	11,12,13 /6		16 /3	14a,14b,15 /6	14c /2	/17
Q.17-22 p 7-9	21 /2	19,22 /6	17,18 /6	20 /2		/16
Q.23-28 p 10-12	23,25,26 /8	27,28 /6			24 /3	/17
Q.29-31 p 13-16	30 /5	31 /5	29 /3			/13
Total	/27	/19	/12	/9	/6	/73



BAULKHAM HILLS HIGH SCHOOL
YEAR 11 Yearly Examination 2020
MATHEMATICS ADVANCED

Mathematics
Section I – Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

**Start
Here** →

- | | |
|--|---|
| 1. A ☐ B ☐ C ☐ D ☐ | 6. A ☐ B ☐ C ☐ D ☐ |
| 2. A ☐ B ☐ C ☐ D ☐ | 7. A ☐ B ☐ C ☐ D ☐ |
| 3. A ☐ B ☐ C ☐ D ☐ | 8. A ☐ B ☐ C ☐ D ☐ |
| 4. A ☐ B ☐ C ☐ D ☐ | 9. A ☐ B ☐ C ☐ D ☐ |
| 5. A ☐ B ☐ C ☐ D ☐ | 10. A ☐ B ☐ C ☐ D ☐ |

2020 Year 11 Yearly Examination

Mathematics Advanced

Section II Answer Booklet

63 marks

Attempt Questions 11 – 31

Allow about 1 hour and 45 minutes for this section

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
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Question 11 (2 marks)

Express $\frac{x+1}{x^2-x} - \frac{x-1}{x^2+x}$ as a single fraction in its simplest form.

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Question 12 (2 marks)

Express $\frac{3\sqrt{2}-2\sqrt{3}}{3\sqrt{2}+2\sqrt{3}}$ in the form $a + b\sqrt{6}$.

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Question 13 (2 marks)

Solve $|x - 4| = 7$.

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Question 14 (6 marks)

Differentiate with respect to x .

(a) $\frac{1}{\sqrt{x^3}}$ 2

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(b) $(1 - 5x)^4$ 2

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(c) $\frac{e^{2x}}{x^2 + 1}$ 2

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Question 15 (2 marks)

Find the equation of the tangent to the curve $y = x^2 + 1$ at the point $(2, 5)$.

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Question 16 (3 marks)

The table below shows a discrete probability distribution for a random variable X , where k is a constant.

x	0	1	2	3	4
$P(X=x)$	0.2	0.1	$2k$	$3k$	0.3

- (a) Find the value of k .

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- (b) Find $P(X \geq 1)$.

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Question 17 (4 marks)

A pair of fair 6-sided die are rolled. The number of sixes showing is represented by a random variable X with the following probability distribution.

x	0	1	2
$p(x)$	$\frac{25}{36}$	$\frac{5}{18}$	$\frac{1}{36}$

- (a) Show that the expected value $E(X)$ is $\frac{1}{3}$. 1

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- (b) Hence, or otherwise, find $\text{Var}(X)$, correct to 2 decimal places. 2

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- (c) Hence, find the standard deviation, correct to 2 decimal places. 1

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Question 18 (2 marks)

Two identical biased coins are tossed together, and the number of heads showing is recorded. After a large number of trials, it is observed that the probability that both coins land showing heads is 0.64. 2

What is the experimental probability that both coins land showing tails?

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Question 19 (3 marks)

Solve $2\sin^2 x - 1 = 0$ for $0 \leq x \leq 2\pi$.

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Question 20 (2 marks)

If $f(x) = x^2 + x$, find its first derivative by using the first principles formula:

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$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

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Question 21 (2 marks)

A circle is given by the equation $x^2 + y^2 + 6x - 8y = 0$.

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Find the centre and radius of this circle.

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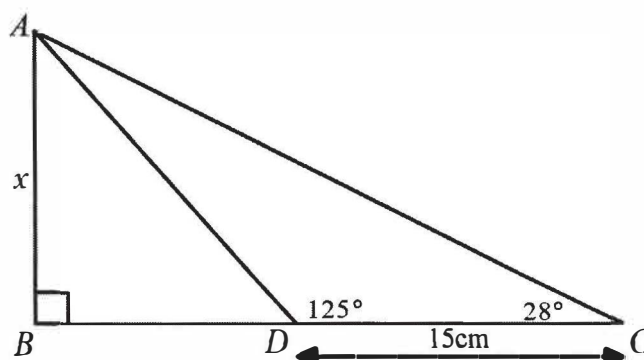
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Question 22 (3 marks)

Find the value of x , correct to 2 decimal places.

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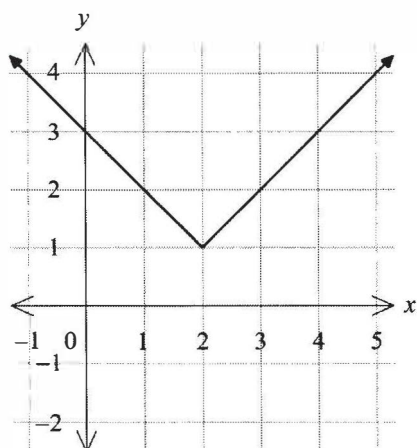
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Question 23 (2 marks)

The graph of the function below is of the form $y = |x + b| + c$, where b and c are constants.

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What are the values of b and c ?

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Question 24 (3 marks)

Solve $4^x - 10(2^x) + 24 = 0$.

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Question 25 (2 marks)

Simplify $\frac{4^m - 1}{2^m - 1}$.

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Question 26 (4 marks)

(a) Show that $\frac{2x+3}{x+2} = 2 - \frac{1}{x+2}$.

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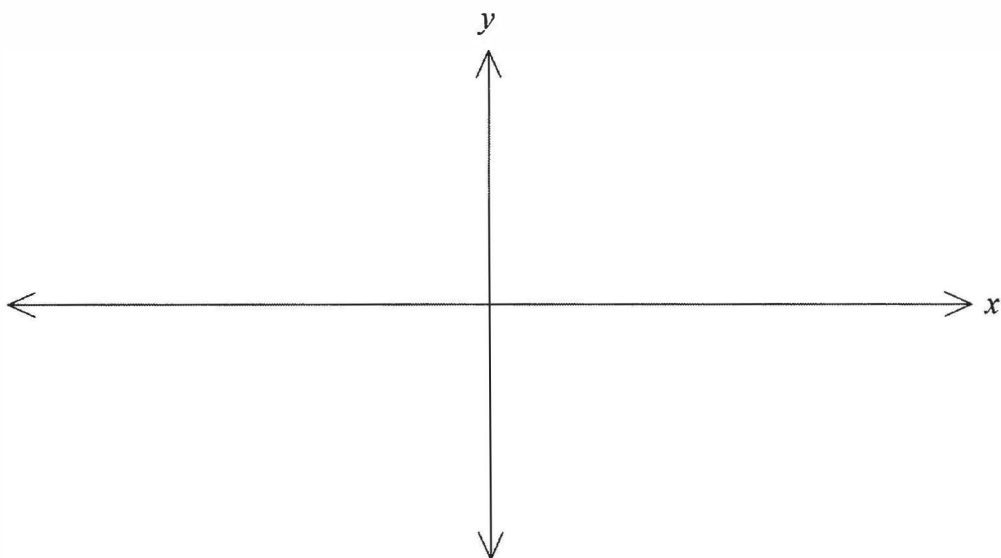
(b) Hence sketch the graph of $y = \frac{2x+3}{x+2}$, showing all intercepts and asymptotes.

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Question 27 (2 marks)

Express $2 \sin(180^\circ - \alpha) + \cos(90^\circ - \alpha)$ in terms of $\sin \alpha$.

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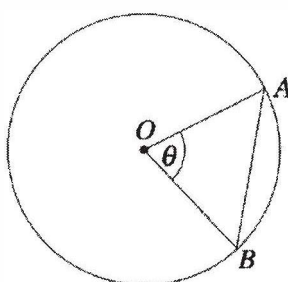
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Question 28 (4 marks)

The diagram shows a circle with centre O and radius 2 centimetres. The points A and B lie on the circumference of the circle and $\angle AOB = \theta$.



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- (a) There are two possible values of θ in radians, for which the area of $\triangle AOB$ is $\sqrt{3}$ square centimetres. One value is $\frac{\pi}{3}$. Find the other value.

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- (b) Suppose that $\theta = \frac{\pi}{3}$. Find the exact area, in square centimetres, of the minor segment bounded by the chord AB and the arc AB .

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Question 29 (3 marks)

For events A and B from a given sample space, $P(A|B) = \frac{1}{5}$ and $P(B|A) = \frac{1}{3}$. Let $P(A \cap B) = p$.

- (a) Find $P(A)$ in terms of p .

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- (b) Find $P(\bar{A} \cap \bar{B})$ in terms of p .

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Question 30 (5 marks)

Consider the polynomial $g(x) = (2 - x)(x + 2)^2$.

- (a) What is the degree of the polynomial? **1**

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- (b) Draw a neat sketch of $g(x) = (2 - x)(x + 2)^2$, showing all x and y intercepts. **2**

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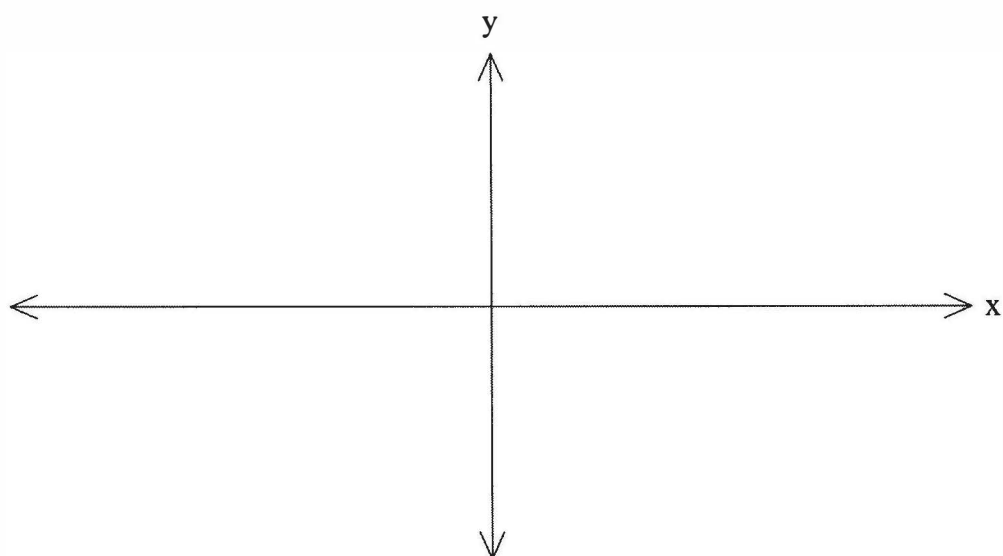
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- (c) For what values of x is $g(x) \leq 0$? **2**

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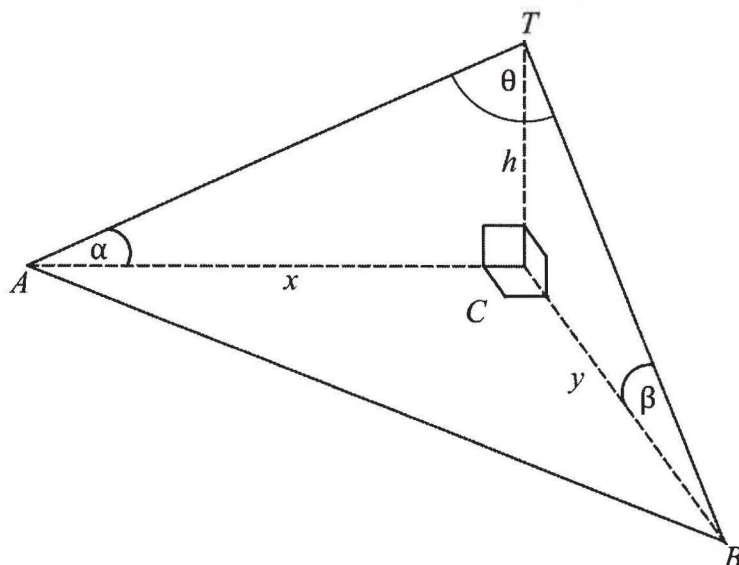
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Question 31 (5 marks)

In the diagram $AC = x$, $BC = y$ and $TC = h$.

$\angle ATB = \theta$, $\angle TAC = \alpha$ and $\angle TBC = \beta$. There are three right angles at C as shown.



- (a) Show that $AT = \sqrt{x^2 + h^2}$ and write a similar expression for BT . 1

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- (b) Show that $\cos\theta = \frac{h^2}{\sqrt{(x^2 + h^2)(y^2 + h^2)}}$. 2

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Question 31 continues on page 19

(c) Hence show that $\sin\alpha\sin\beta = \cos\theta$.

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End of paper



Solutions

**BAULKHAM HILLS
HIGH SCHOOL**

2020

**YEAR 11 YEARLY
EXAMINATION**

Mathematics Advanced

**General
Instructions**

- Reading time – 10 minutes
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**Total marks:
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Section I – 10 marks (pages 2–5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 63 marks (Answer Booklet pages 4–16)

- Attempt Questions 11 – 31
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

1

What is the value of $\sqrt{\frac{28-\pi^2}{5.3}}$ correct to 2 significant figures?

(A.) 1.8

B. 1.84

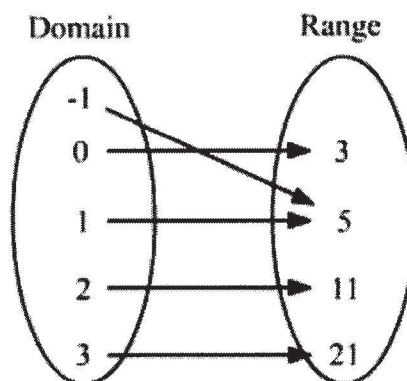
C. 1.85

D. 1.9

$$\approx 1.849548\dots$$

$$\approx 1.8$$

2 What type of relation is shown?



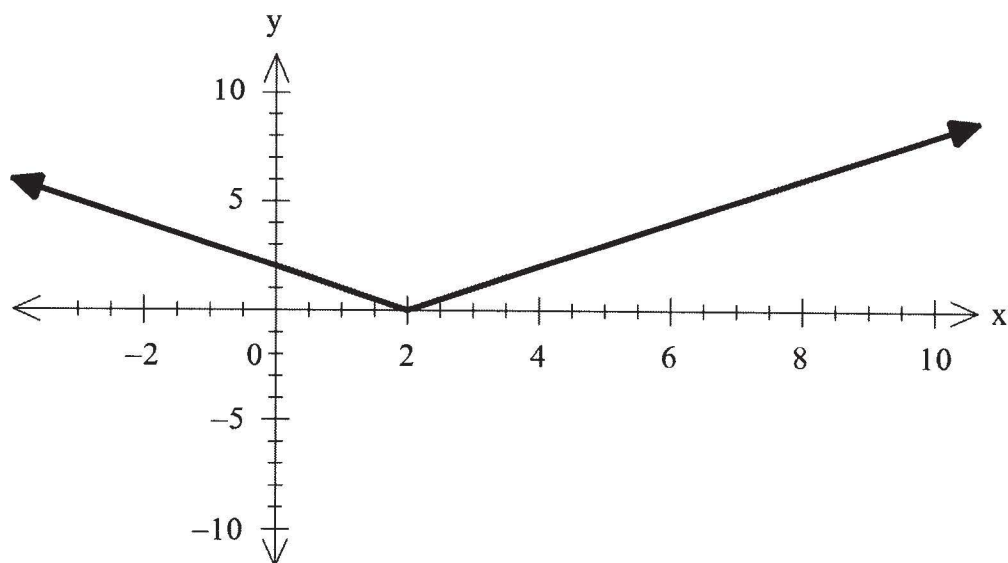
A. One to one

B. One to many

(C.) Many to one

D. Many to many

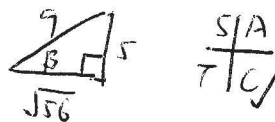
- 3 Which of the following statements is true about the following function?



- A. The function is neither continuous nor differentiable at $x = 2$.
- ☒ B. The function is continuous but not differentiable at $x = 2$.
- C. The function is differentiable but not continuous at $x = 2$.
- D. The function is both continuous and differentiable at $x = 2$.
- 4 Given that $\operatorname{cosec} \beta = -\frac{9}{5}$ and that $\tan \beta < 0$, what is the value of $\cos \beta$?

- A. $-\frac{9}{\sqrt{106}}$
- B. $-\frac{\sqrt{56}}{9}$
- ☒ C. $\frac{\sqrt{56}}{9}$
- D. $\frac{9}{\sqrt{106}}$

$$\sin \beta = -\frac{5}{9}$$



$$\therefore \cos \beta = \frac{\sqrt{56}}{9}$$

5 If $\log_b x = 3$ and $\log_b y = 5$, what is the value of $\log_b(xy^2)$?

A. 8

☒ B. 13

C. 28

D. 75

$$\begin{aligned} &= \log_b x + 2 \log_b y \\ &= 3 + 2(5) \\ &= 13 \end{aligned}$$

6 What is the minimum value of the function $f(x) = 2x^2 - 4x + 1$?

A. -2

☒ B. -1

C. 1

D. 2

$$\begin{aligned} \text{axis of symmetry: } &= \frac{-b}{2a} \\ &= \frac{4}{4} \\ &= 1 \\ \therefore f(1) &= 2(1)^2 - 4(1) + 1 \\ &= -1 \end{aligned}$$

7 What is the natural domain of $f(x) = \frac{1}{\sqrt{x^2-9}}$?

☒ A. $x < -3$ and $x > 3$

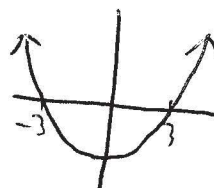
B. $x \leq -3$ and $x \geq 3$

C. $-3 < x < 3$

D. $-3 \leq x \leq 3$

$$x^2 - 9 > 0$$

$$(x+3)(x-3) > 0$$



8 Which of the following is an equivalent expression for $\frac{(1+\cos x)(1-\cos x)}{\sin^2 x}$?

A. $\tan^2 x$

B. $\cot^2 x$

C. $\operatorname{cosec}^2 x$

☒ D. 1

$$= \frac{1 - \cos^2 x}{\sin^2 x}$$

$$= \frac{\sin^2 x}{\sin^2 x}$$

$$= 1$$

9 If $f(x) = x^2 - 2x$ and $g(x) = \sqrt{x+1}$, what is $g(f(x))$?

A. $(\sqrt{x+1})(x^2 - 2x)$

B. $x + 1 - 2\sqrt{x+1}$

C. $\sqrt{x^2 - x + 1}$

☒ D. $|x - 1|$

$$g(f(x)) = \sqrt{(x^2 - 2x) + 1}$$

$$= \sqrt{(x-1)^2}$$

$$= |x-1|$$

10 Which of the following functions is **not** even?

A. $f(x) = \sqrt{x^2 - 1}$

B. $f(x) = \frac{1}{1+x^2}$

☒ C. $f(x) = \operatorname{cosec} x$

D. $f(x) = \cos x$

End of Section I

2020

Year 11 Yearly Examination

Mathematics Advanced

Section II Answer Booklet

63 marks**Attempt Questions 11 – 31****Allow about 1 hour and 45 minutes for this section**

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
-

Question 11 (2 marks)

Express $\frac{x+1}{x^2-x} - \frac{x-1}{x^2+x}$ as a single fraction in its simplest form.

2

$$\begin{aligned}
 &= \frac{x+1}{x(x-1)} - \frac{x-1}{x(x+1)} \\
 &= \frac{(x+1)(x+1) - (x-1)(x-1)}{x(x-1)(x+1)} \\
 &= \frac{x^2+2x+1 - x^2+2x-1}{x(x-1)(x+1)} \\
 &= \frac{4x}{x(x-1)(x+1)} \\
 &= \frac{4}{(x+1)(x-1)}
 \end{aligned}$$

2MK: Provides correct solution.
1MK: Obtains correct expression with the same denominator

Question 12 (2 marks)

Express $\frac{3\sqrt{2}-2\sqrt{3}}{3\sqrt{2}+2\sqrt{3}}$ in the form $a + b\sqrt{6}$.

2

$$\begin{aligned}
 \frac{3\sqrt{2}-2\sqrt{3}}{3\sqrt{2}+2\sqrt{3}} \times \frac{3\sqrt{2}-2\sqrt{3}}{3\sqrt{2}-2\sqrt{3}} &= \frac{18 - 2(6\sqrt{6}) + 12}{18 - 12} \\
 &= \frac{30 - 12\sqrt{6}}{6} \\
 &= 5 - 2\sqrt{6}
 \end{aligned}$$

2MKs Provides correct solution.
1MK: Attempts to rationalise by using the conjugate

Question 13 (2 marks)

Solve $|x - 4| = 7$.

2

$$\begin{aligned}
 x-4 &= 7 & x-4 &= -7 \\
 x &= 11 & x &= -3 \\
 \therefore x &= 11 \text{ or } -3
 \end{aligned}$$

2MKs: Provides correct solution.
1MK: Correctly solves one case.

Question 14 (6 marks)

Differentiate with respect to x .

(a) $\frac{1}{\sqrt{x^3}}$

2

$$\frac{d}{dx} \left(\frac{1}{\sqrt{x^3}} \right) = \frac{d}{dx} \left(x^{-\frac{3}{2}} \right)$$

2MK: Provides correct solution.

$$= -\frac{3}{2} x^{-\frac{5}{2}} \text{ or } \frac{-3}{2\sqrt{x^5}}$$

1MK: Obtains correct index

form for the expression.

(b) $(1-5x)^4$

2

$$\frac{d}{dx} (1-5x)^4 = 4(1-5x)^3 \cdot -5$$

2MK: Provides correct solution

$$= -20(1-5x)^3$$

1MK: Attempts to use the chain rule

-Obtains $4(1-5x)^3$

(c) $\frac{e^{2x}}{x^2+1}$

2

$$u = e^{2x}$$

$$v = x^2+1$$

$$u' = 2e^{2x}$$

$$v' = 2x$$

2MK: Provides correct solution

1MK: Attempts to use

$$\therefore \frac{dy}{dx} = \frac{(x^2+1)(2e^{2x}) - e^{2x}(2x)}{(x^2+1)^2}$$

the quotient rule

$$= \frac{2e^{2x}(x^2-x+1)}{(x^2+1)^2}$$

Question 15 (2 marks)

Find the equation of the tangent to the curve $y = x^2 + 1$ at the point (2, 5).

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$y' = 2x$ 2Mk: Provides correct solution

When $x=2$, $y' = 2(2)$ 1Mk: Obtains correct gradient

$= 4$ at $x=2$ is 4.

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$\therefore$ Eqn of tangent: $y-5 = 4(x-2)$

$y-5 = 4x-8$

$\therefore 4x-y-3=0$ or $y=4x-3$

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Question 16 (3 marks)

The table below shows a discrete probability distribution for a random variable X , where k is a constant.

x	0	1	2	3	4
$P(X=x)$	0.2	0.1	$2k$	$3k$	0.3

(a) Find the value of k .

2

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$0.2 + 0.1 + 2k + 3k + 0.3 = 1$ 2Mk: Provides correct solution

$\therefore 5k = 0.4$ 1Mk: Obtains a correct

$k = 0.08$ probability statement to

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solve for k .

(b) Find $P(X \geq 1)$.

1

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$1 - 0.2 = 0.8$ 1Mk: Provides correct solution

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Question 17 (4 marks)

A pair of fair 6-sided die are rolled. The number of sixes showing is represented by a random variable X with the following probability distribution.

x	0	1	2
$p(x)$	$\frac{25}{36}$	$\frac{5}{18}$	$\frac{1}{36}$

- (a) Show that the expected value $E(X)$ is $\frac{1}{3}$. 1
- $E(X) = 2 \times \frac{1}{36} + 1 \times \frac{5}{18} + 0 = \frac{1}{3}$ // 1mk: Provides correct proof.
- (b) Find $\text{Var}(X)$, correct to 2 decimal places. 2
- $\text{Var}(X) = (4 \times \frac{1}{36} + 1 \times \frac{5}{18} + 0) - (\frac{1}{3})^2$
- $= 0.2777...$ 2mk: Provides correct solution.
- ≈ 0.28 1mk: Attempts to find variance using a correct method.
- (c) Hence, find the standard deviation, correct to 2 decimal places. 1
- $\sqrt{0.2777} = 0.527...$
- ≈ 0.53 1mk: Provides correct solution.

Question 18 (2 marks)

Two identical biased coins are tossed together, and the number of heads showing is recorded. After a large number of trials, it is observed that the probability that both coins land showing heads is 0.64. 2

What is the experimental probability that both coins land showing tails?

$P(H) \times P(H) = 0.64$

$\therefore P(H) = 0.8$ 2mk: Provides correct solution.

$\therefore P(T) = 0.2$ 1mk: Obtains correct $P(\text{Head})$

$P(T) \times P(T) = 0.2 \times 0.2 = 0.04$

Question 19 (3 marks)

Solve $2\sin^2 x - 1 = 0$ for $0 \leq x \leq 2\pi$.

3

$$\sin^2 x = \frac{1}{2}$$

$$\sin x = \pm \sqrt{\frac{1}{2}}$$

$$= \pm \frac{1}{\sqrt{2}}$$

3MK: Provides 4 correct solutions.

2MK: Provides 2 correct solutions

$\therefore x$ is in all 4 quadrants

1MK: Obtains an equation for $\sin x$.

$$\therefore x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

$$\text{ie } \sin x = \pm \frac{1}{\sqrt{2}}$$

Question 20 (2 marks)

If $f(x) = x^2 + x$, find its first derivative by using the first principles formula:

2

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 + (x+h) - (x^2 + x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 + (x+h) - x^2 - x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + x + h - x^2 - x}{h}$$

2MK: Provides correct solution.

$$= \lim_{h \rightarrow 0} \frac{h(2x + h + 1)}{h}$$

1MK: Substitutes correctly into the first principles formula

$$= 2x + (0) + 1$$

$$= 2x + 1$$

Note: limit notation enforced.

Question 21 (2 marks)

A circle is given by the equation $x^2 + y^2 + 6x - 8y = 0$.

2

Find the centre and radius of this circle.

$$(x+3)^2 + (y-4)^2 = 9+16$$

2mk: Provides correct centre and radius.

$$\therefore (x+3)^2 + (y-4)^2 = 25$$

1mk: Provides correct centre or radius.

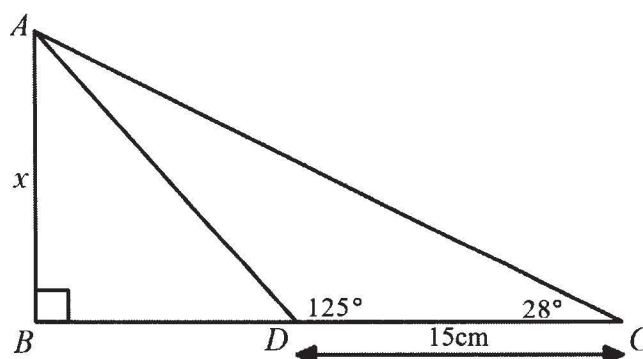
$\therefore$ Centre $(-3, 4)$.

& Radius is 5.

Question 22 (3 marks)

Find the value of x , correct to 2 decimal places.

3



$$\text{In } \triangle ADC: \frac{AD}{\sin 28} = \frac{15}{\sin 27}$$

3mk: Provides correct solution.

$$\therefore AD = \frac{15 \sin 28}{\sin 27}$$

2mk: Obtains a correct trig

$$\text{In } \triangle ABD: \frac{x}{AD} = \sin 55$$

ratio expression involving $\sin$.

$$x = \left(\frac{15 \sin 28}{\sin 27} \right) \sin 55$$

1mk: Obtains a correct sine

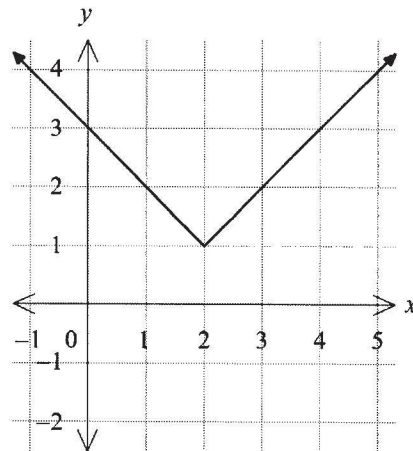
$$= 12.706...$$

rule equation.

$$\hat{=} 12.71 \text{ cm}$$

Question 23 (2 marks)

The graph of the function below is of the form $y = |x + b| + c$, where b and c are constants as shown.

2

What are the values of b and c ?

.....
 $b = -2, c = 1$

2MK: Provides two correct solutions
 1MK: Identifies one correct constant.

Question 24 (3 marks)

Solve $4^x - 10(2^x) + 24 = 0$.

3

.....
 Let $u = 2^x$

 $\therefore u^2 - 10u + 24 = 0$
 $(u - 4)(u - 6) = 0$
 $\therefore u = 4 \text{ or } 6$
 $\therefore 2^x = 4$ $2^x = 6$
 $x = 2$ $x = \log_2 6$

3MK: Correctly finds both solutions.
 2MK: Correctly finds one solution.
 1MK: - Correctly obtains a factorised expression
 - Correctly substitutes into the quadratic formula

Question 25 (2 marks)

Simplify $\frac{4^m - 1}{2^m - 1}$.

2

$$= \frac{(2^m)^2 - 1}{2^m - 1}$$

$$2^m - 1$$

2mk: Provides correct solution.

$$= \frac{(2^m + 1)(2^m - 1)}{2^m - 1}$$

1mk: Rewrites 4^m as $(2^m)^2$

$$= 2^m + 1$$

Question 26 (4 marks)

(a) Show that $\frac{2x+3}{x+2} = 2 - \frac{1}{x+2}$.

1

$$\text{RHS} = \frac{2(x+2) - 1}{x+2}$$

$$= \frac{2x + 4 - 1}{x+2}$$

1mk: Provides correct proof.

$$= \frac{2x+3}{x+2}$$

= LHS as required

(b) Hence sketch the graph of $y = \frac{2x+3}{x+2}$, showing all intercepts and asymptotes.

3

$$\text{When } x=0, y = \frac{3}{2}$$

$$1 = 2x + 4$$

3mk: Provides correct sketch.

$$\text{When } y=0, 0 = 2 - \frac{1}{x+2}$$

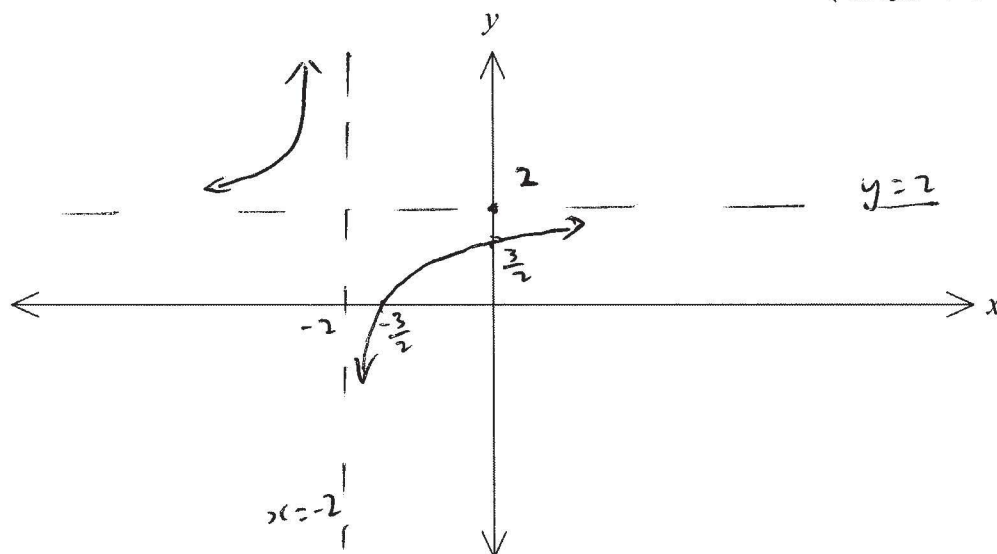
$$-3 = 2x$$

2mk: Provides curve with correct intercepts and asymptotes

$$\frac{1}{x+2} = 2$$

$$x = -\frac{3}{2}$$

1mk: Provides curve with correct intercepts or asymptotes.



Question 27 (2 marks)

Express $2 \sin(180^\circ - \alpha) + \cos(90^\circ - \alpha)$ in terms of $\sin \alpha$.

2

$$= 2 \sin \alpha + \sin \alpha$$

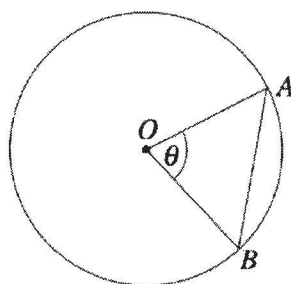
2mk: Provides correct solution.

$$= 3 \sin \alpha$$

1mk: Simplifies one correct identity

Question 28 (4 marks)

The diagram shows a circle with centre O and radius 2 centimetres. The points A and B lie on the circumference of the circle and $\angle AOB = \theta$.



NOT TO SCALE

- (a) There are two possible values of θ in radians, for which the area of $\triangle AOB$ is $\sqrt{3}$ square centimetres. One value is $\frac{\pi}{3}$. Find the other value. 2

$$A = \frac{1}{2} ab \sin C$$

2mk: Provides correct solution.

$$\sqrt{3} = \frac{1}{2} (2)(2) \sin \theta$$

1mk: Obtains correct equation for area of a triangle with substitution.

$$\frac{\sqrt{3}}{2} = \sin \theta$$

$$\therefore \theta = \frac{\pi}{3} \text{ and } \frac{2\pi}{3}$$

- Bold answer

- (b) Suppose that $\theta = \frac{\pi}{3}$. Find the exact area, in square centimetres, of the minor segment bounded by the chord AB and the arc AB . 2

$$A_{\text{segment}} = A_{\text{sector}} - A_{\text{triangle}}$$

2mk: Provides correct solution in exact form.

$$= \frac{1}{2} r^2 \theta - \sqrt{3}$$

1mk: Obtains correct equation for area of minor segment with substitution.

$$= \frac{1}{2} (2)^2 \left(\frac{\pi}{3}\right) - \sqrt{3}$$

$$= \frac{2\pi}{3} - \sqrt{3} \text{ cm}^2$$

Question 29 (3 marks)

For events A and B from a given sample space, $P(A|B) = \frac{1}{5}$ and $P(B|A) = \frac{1}{3}$. Let $P(A \cap B) = p$.

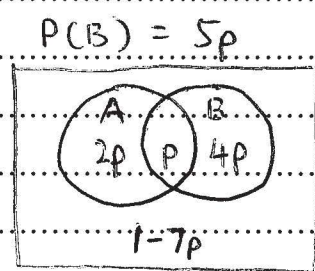
- (a) Find $P(A)$ in terms of p .

1

$$\begin{aligned}
 P(A) &= \frac{P(A \cap B)}{P(B|A)} && \text{1Mk: Provides correct solution} \\
 &= \frac{p}{(\frac{1}{3})} \\
 &= 3p
 \end{aligned}$$

- (b) Find $P(\bar{A} \cap \bar{B})$ in terms of p .

2



2Mk: Provides correct solution.

1Mk: Obtains $P(B) = 5p$
or equivalent merit.

$$\therefore P(\bar{A} \cap \bar{B}) = 1 - 7p$$

Question 30 (5 marks)

Consider the polynomial $g(x) = (2 - x)(x + 2)^2$.

- (a) What is the degree of the polynomial?

1

3

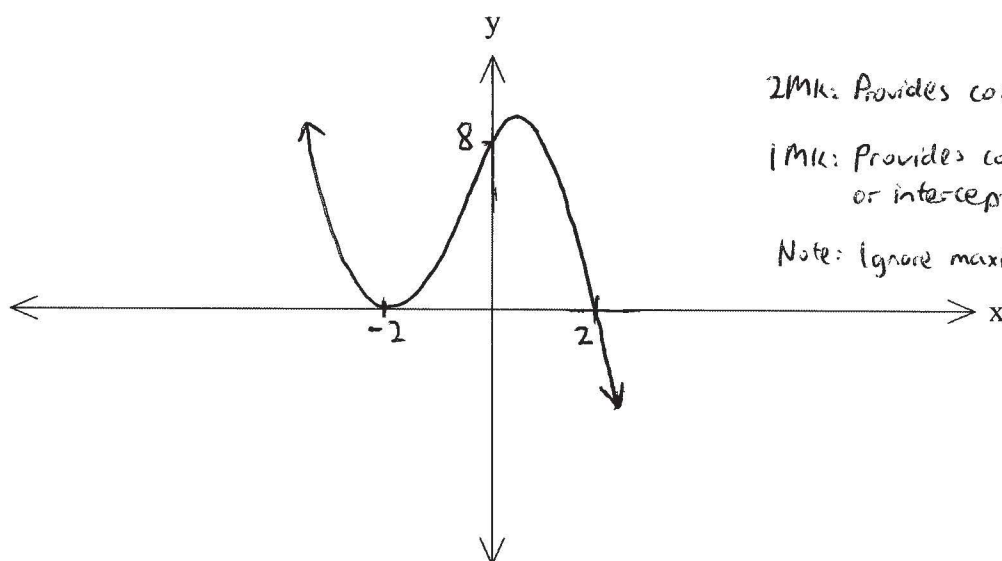
1mk: Provides correct answer.

- (b) Draw a neat sketch of $g(x) = (2 - x)(x + 2)^2$, showing all x and y intercepts.

2

x-int: $x = 2, -2$

y-int: $y = (2)(2)^2$
 $= 8$



2mk: Provides correct sketch.

1mk: Provides correct shape or intercepts.

Note: Ignore maximum turning point.

- (c) For what values of x is $g(x) \leq 0$?

2

From graph: $x = -2, x \geq 2$

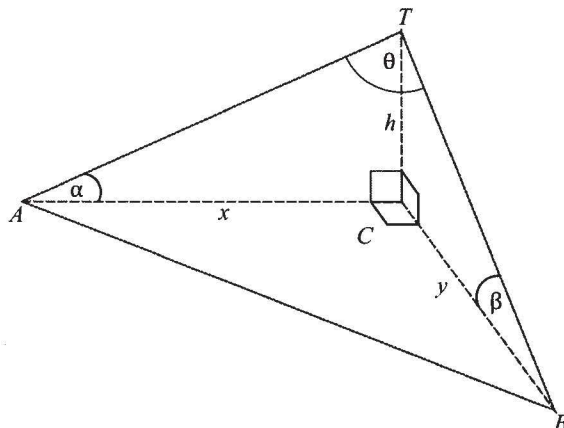
2mk: Provides correct solution.

1mk: Obtains part of the correct x -values.

Question 31 (5 marks)

In the diagram $AC = x$, $BC = y$ and $TC = h$.

$\angle ATB = \theta$, $\angle TAC = \alpha$ and $\angle TBC = \beta$. There are three right angles at C as shown.



- (a) Show that $AT = \sqrt{x^2 + h^2}$ and write a similar expression for BT .

1

In ΔATC : $AT^2 = x^2 + h^2$ 1MK: Provides correct proof.
 $\therefore AT = \sqrt{x^2 + h^2}$ & $BT = \sqrt{y^2 + h^2}$

- (b) Show that $\cos \theta = \frac{h^2}{\sqrt{(x^2 + h^2)(y^2 + h^2)}}$.

2

In ΔATB : $(AB)^2 = (AT)^2 + (BT)^2 - 2(AT)(BT) \cos \theta$
 $x^2 + y^2 = (x^2 + h^2) + (y^2 + h^2) - 2\sqrt{x^2 + h^2} \sqrt{y^2 + h^2} \cos \theta$
 $\therefore 0 = 2h^2 - 2\sqrt{(x^2 + h^2)(y^2 + h^2)} \cos \theta$
 $\cos \theta = \frac{h^2}{\sqrt{(x^2 + h^2)(y^2 + h^2)}}$ as required. 2MK: Provides correct proof.
1MK: Obtains correct equation for cosine rule in ΔATB .

- (c) Hence show that $\sin \alpha \sin \beta = \cos \theta$.

2

In ΔATC : $\sin \alpha = \frac{h}{AT}$ & In ΔBTC : $\sin \beta = \frac{h}{BT}$
 $= \frac{h}{\sqrt{x^2 + h^2}}$ (from (i)) 1MK: Provides correct proof.
 $= \frac{h}{\sqrt{y^2 + h^2}}$ (from (i))
 $\therefore \sin \alpha \sin \beta = \frac{h^2}{\sqrt{x^2 + h^2} \sqrt{y^2 + h^2}}$
 $= \frac{h^2}{\sqrt{(x^2 + h^2)(y^2 + h^2)}}$ 1MK: Obtains $\sin \alpha$ or $\sin \beta$ correctly.
 $= \cos \theta$

End of paper